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Assignment - 2

section : A

ID : 2230107/3

Q-1

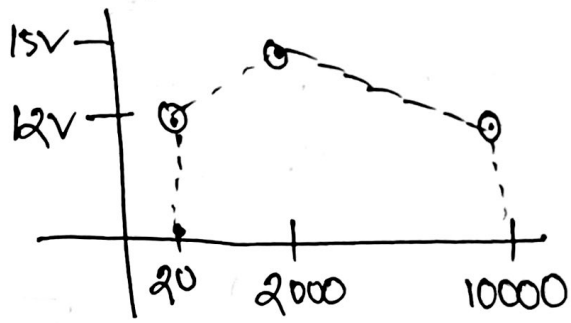
$f_1 = 20$

$f_2 = 1/5\times10^{-4} = 2000$

$f_3 = 10000$

$\therefore$ Bandwidth $= (10000 - 20) = 9980\,Hz$

Diagram



Q-2

$SNR = 10^{\frac{20}{10}}$

$= 100$

$B = 40\,kHz$

$= 40000\,Hz$

$C = 40000 \log_2(1+100)$

$= 40000 \times 6.6$

$= 266400\ bits^{-1}$

$= 266\ kbps$

Q-3

$B = 3 \times 10^6$ Hz

$W = 0.02$ w

$S = 41.4$ w

$$SNR = \frac{41.4}{0.02} = 45$$

$$C = (3 \times 10^6) \log_2 (1 + 45)$$

$$= 16.57 \text{ Mbps}$$

20% lower, So, $B = C \times (1 - 0.20)$

$$= 16.57 \times 0.8 \text{ Mbps}$$

$$= 13.256 \text{ Mbps}$$

$$B = 2B \log_2 (L)$$

$$13.256 \times 10^6 = (6 \times 10^6) \times \log_2 L$$

$$\log_2 (L) = 2.2093$$

$$L = 4.62$$

$\therefore$ Bit rate = 13.256

Signal level = 5

Q-4

$$B = 2 \times B \times \log_2(L)$$

$$= 2 \times (20 \times 10^6) \times 4$$

$$= 160 \text{ mbps}$$

$B = 20 \text{ mHz} = 20 \times 10^6$

$\log_2(L) = 4$

90% of the theoritical upper capacity

$$C' = 0.90 \times C$$

$$C = \frac{160 \text{ mbps}}{0.90}$$

$$= 177.78 \text{ mbps}$$

Q-6

a) 01011000 (Bipolar AMI)

b) 11011001 (NRZ-L)

c) 01001110 (MLT-3)

d) 00111011 (Manchester)

e) 01011001 (Unipolar)

f) 00110101 (NRZ-I)

Q-5

Page = 8

Line = 25

Char = 80

Bit per char = 8

$$\text{total} = (8 \times 25 \times 80 \times 8)$$
$$= 128{,}000 \text{ bits}$$

$R = 4\,Mbps$
$= 4 \times 10^6$ bps

$$\text{transmission delay} = \frac{128000}{4 \times 10^{-6}} \text{ bps}$$

$$\text{propagation delay} = \frac{6 \times 10^6}{2.8 \times 10^8}$$
$$= 0.021428 \text{ s}$$
$$= 21.43 \text{ ms}$$

$d = 6000\,km$
$= 6 \times 10^6\,m$

$v = 2.8 \times 10^8\,m/s$

Processing delay: 2×6 ms
$= 12$

$$\therefore \text{total} = (32 + 21.43 + 12)$$
$$= 65.43 \text{ ms}$$

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(i) 101111110101001

5B		4B
10111	—	1011
11101	—	1111
01001	—	0001

original bit stream = 101111110001

(ii) no of bits = 12 bits (in 4B encoding)

no of bits = 15 bits (in 5B)

redundant = 15 − 12

= 3

Percentage $\frac{3 \times 100}{12}$

= 25%